MATH& 142 — Midterm Examination

Modules 1–5: Trigonometry and Modeling

Name: _____ Date: _____

Instructions. Show work and explain your reasoning. Give exact answers unless the question requests a decimal approximation. Angles are in radians unless marked with a degree symbol. A non-CAS scientific calculator is permitted. Total: 100 points.

1. **Angles and arc length (10 points).** Convert 225° to radians. Then calculate the arc length traced by a point on a wheel of radius 6 cm through an angle of $5\pi/6$ radians.

2. **Unit circle (10 points).** Determine exactly $\sin(7\pi/6)$, $\cos(7\pi/6)$, and $\tan(7\pi/6)$. State the reference angle and justify the signs.

3. **Six trigonometric functions (10 points).** Suppose $\sin\theta = 3/5$ and θ lies in Quadrant II. Find $\cos\theta$, $\sec\theta$, and $\cot\theta$ exactly.

4. **Graph transformations (10 points).** For $f(x) = -3\cos(2(x - \pi/4)) + 1$, determine amplitude, period, phase shift, midline, and range. Explain the meaning of the negative amplitude coefficient.

5. **Modeling a rotating wheel (10 points).** A rider starts at the rightmost point of a counterclockwise wheel of radius 9 m, centered 11 m above ground. The wheel completes a revolution every 36 seconds. Write a model for the rider's height $h(t)$, and calculate the height at $t = 9$ seconds.

6. **Inverse trigonometry (10 points).** Evaluate exactly $\arcsin(\sin(5\pi/6))$. Then find $\cos(\arcsin(3/5))$ exactly and explain the principal-value restriction.

7. **Trigonometric identity (10 points).** Prove $\dfrac{1-\cos^2 x}{\sin x} = \sin x$ and state the set of inputs for which the displayed equality is defined.

8. **Solving an equation (10 points).** Find *all* solutions of $\sin(2x) = \sqrt{3}/2$ on $[0, 2\pi)$. Show how you account for multiple cycles.

9. **Law of Sines (10 points).** In triangle ABC, $A = 40^\circ$, $B = 65^\circ$, and $a = 12$ cm. Determine C and side b. Round the side length to two decimal places.

10. **Law of Cosines (10 points).** Two sides of a triangle are 8 m and 11 m, with included angle 60°. Find the third side exactly and approximately to two decimal places.